

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2135

Drewry 100 kV
March 20, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2135. The Interconnection Customer proposes a 122.0 MW battery energy storage facility interconnecting at the Drewry 100 kV substation in the Indiana area. The request was received on July 15, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 550 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$374,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
EAST ASTORIA - COLFAX 100 KV	GASTON RD - TABLEMT 100 KV	102.5	285	47.6
RICLAND - NAPOLEON PEAKING STA 230 KV	RICHLAND - TROY ENERGY LLC 500 KV L	101.3	1,356	11.9
GOULD - STA S VAN NUYS 230 KV	TROY ENERGY LLC - RICHLAND 500 KV	100.3	811	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EAST ASTORIA - COLFAX 100 KV	Reconductor line	\$1,146,000	19% (\$219,000)
RICLAND - NAPOLEON PEAKING STA 230 KV	Reconductor line	\$13,963,000	1% (\$112,000)
STA S VAN NUYS - GOULD 230 KV	Reconductor line	\$4,705,000	1% (\$43,000)
	Total allocated to project		\$374,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.